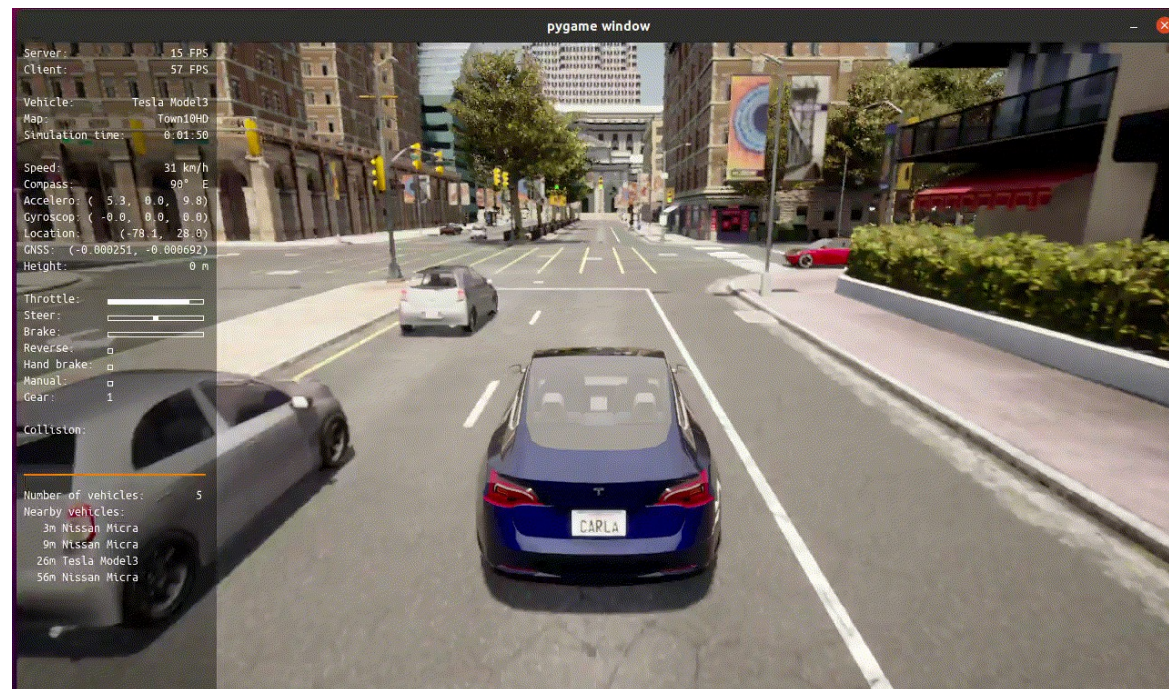
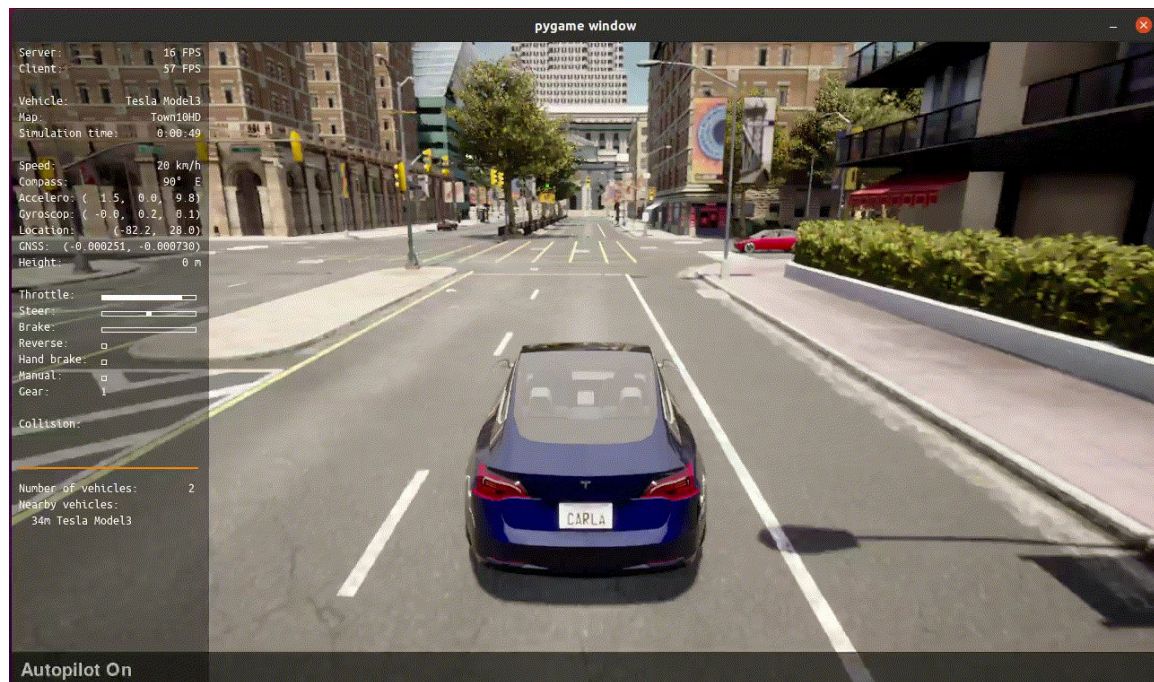


基于 generator-judger 的树状仿真代码生成 agent
怎么翻译自然语言到仿真场景？

Q1：怎么选初始点？

Q2：怎么选择控制代码

Q3：怎么输出适配于当前格式（xml）的生成方式



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9.1.1. Mechanics

Point-kinetics in a fixed coordinate system

Definitions

The position $\vec{r}$, the velocity $\vec{v}$ and the acceleration $\vec{a}$ are defined by: $\vec{r} = (x, y, z)$, $\vec{v} = (\dot{x}, \dot{y}, \dot{z})$, $\vec{a} = (\ddot{x}, \ddot{y}, \ddot{z})$. The following holds:

$$s(t) = s_0 + \int |\vec{v}(t)| dt; \quad \vec{r}(t) = \vec{r}_0 + \int \vec{v}(t) dt; \quad \vec{v}(t) = \vec{v}_0 + \int \vec{a}(t) dt \quad (9.1.1.1)$$

When the acceleration is constant this gives: $\vec{v}(t) = \vec{v}_0 + \vec{a}t$ and $s(t) = s_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$.

For the unit vectors in a direction $\perp$ to the orbit $\vec{e}_t$ and parallel to it $\vec{e}_n$:

$$\vec{e}_t = \frac{\vec{v}}{|\vec{v}|} = \frac{d\vec{r}}{ds} \quad \vec{e}_t = \frac{v}{\rho} \vec{e}_n \quad \vec{e}_n = \frac{\dot{\vec{e}}_t}{|\dot{\vec{e}}_t|} \quad (9.1.1.2)$$

For the curvature k and the radius of curvature ρ :

$$\vec{k} = \frac{d\vec{e}_t}{ds} = \frac{d^2\vec{r}}{ds^2} = \left| \frac{d\varphi}{ds} \right|; \quad \rho = \frac{1}{|k|} \quad (9.1.1.3)$$

Polar coordinates

Polar coordinates are defined by: $x = r \cos(\theta)$, $y = r \sin(\theta)$. So, for the unit coordinate vectors: $\vec{e}_r = \hat{\theta} \vec{e}_\theta$, $\vec{e}_\theta = -\hat{\theta} \vec{e}_r$.

The velocity and the acceleration are derived from:

$$\vec{r} = r \vec{e}_r, \quad \vec{v} = \dot{r} \vec{e}_r + r \dot{\theta} \vec{e}_\theta, \quad \vec{a} = (\ddot{r} - r \dot{\theta}^2) \vec{e}_r + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \vec{e}_\theta \quad (9.1.1.4)$$

Relative motion

For the motion of a point D w.r.t. a point Q: $\vec{r}_D = \vec{r}_Q + \frac{\vec{\omega} \times \vec{r}_Q}{\omega^2}$ with $\vec{QD} = \vec{r}_D - \vec{r}_Q$ and $\omega = \dot{\theta}$.

Further a prime on a symbol $\alpha = \theta'$ means that the quantity is defined in a moving system of coordinates. In a moving system: $\vec{v} = \vec{v}_Q + \vec{v}' + \vec{\omega} \times \vec{r}'$ and $\vec{a} = \vec{a}_Q + \vec{a}' + \vec{\alpha} \times \vec{r}' + 2\vec{\omega} \times \vec{v}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}')$ with $\vec{\omega} \times (\vec{\omega} \times \vec{r}') = -\omega^2 \vec{r}'_n$.

Point-dynamics in a fixed coordinate system

Force, (angular) momentum and energy

Newton's 2nd law connects the force on an object and the resulting acceleration of the object where the momentum is given by $\vec{p} = m\vec{v}$:

$$\vec{F}(\vec{r}, \vec{v}, t) = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m \frac{d\vec{v}}{dt} + \vec{v} \frac{dm}{dt} \stackrel{m=\text{const}}{=} m\vec{a}$$

Newton's 3rd law is given by: $\vec{F}_{\text{action}} = -\vec{F}_{\text{reaction}}$.

For the power P : $P = \dot{W} = \vec{F} \cdot \vec{v}$. For the total energy W , the kinetic energy T and the potential energy U : $W = T + U$; $\dot{T} = -\dot{U}$ with $T = \frac{1}{2} m v^2$.

The kick $\vec{S}$ is given by: $\vec{S} = \Delta \vec{p} = \int \vec{F} dt$

The work A , delivered by a force, is $A = \int \vec{F} \cdot d\vec{s} = \int F \cos(\alpha) ds$

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$$Q(x, x', t) = \frac{1}{2} [\delta(x - x' - vt) + \delta(x - x' + vt)]$$

$$P(x, x', t) = \begin{cases} \frac{1}{2v} & \text{if } |x - x'| < vt \\ 0 & \text{if } |x - x'| > vt \end{cases}$$

Further the relation: $Q(x, x', t) = \frac{\partial P(x, x', t)}{\partial t}$ holds.

Waveguides and resonating cavities

The boundary conditions for a perfect conductor can be derived from Maxwell's equations. If $\vec{n}$ is a unit vector $\perp$ the surface, pointing from 1 to 2, and $\vec{K}$ is a surface current density, then:

$$\begin{aligned} \vec{n} \cdot (\vec{D}_2 - \vec{D}_1) &= \sigma & \vec{n} \times (\vec{E}_2 - \vec{E}_1) &= 0 \\ \vec{n} \cdot (\vec{B}_2 - \vec{B}_1) &= 0 & \vec{n} \times (\vec{H}_2 - \vec{H}_1) &= \vec{K} \end{aligned} \quad (9.1.3.12)$$

In a waveguide because of the cylindrical symmetry: $\vec{E}(\vec{x}, t) = \vec{E}(x, y)e^{i(kz - \omega t)}$ and $\vec{B}(\vec{x}, t) = \vec{B}(x, y)e^{i(kz - \omega t)}$ holds. From this one can now deduce that, if B_z and E_z are not $\equiv 0$:

$$\begin{aligned} B_x &= \frac{i}{\epsilon\mu\omega^2 - k^2} \left(k \frac{\partial B_z}{\partial x} - \epsilon\mu\omega \frac{\partial E_z}{\partial y} \right) & B_y &= \frac{i}{\epsilon\mu\omega^2 - k^2} \left(k \frac{\partial B_z}{\partial y} + \epsilon\mu\omega \frac{\partial E_z}{\partial x} \right) \\ E_x &= \frac{i}{\epsilon\mu\omega^2 - k^2} \left(k \frac{\partial E_z}{\partial x} + \epsilon\mu\omega \frac{\partial B_z}{\partial y} \right) & E_y &= \frac{i}{\epsilon\mu\omega^2 - k^2} \left(k \frac{\partial E_z}{\partial y} - \epsilon\mu\omega \frac{\partial B_z}{\partial x} \right) \end{aligned} \quad (9.1.3.13)$$

Now one can distinguish between three cases:

1. $B_z \equiv 0$: the Transverse Magnetic Modes (TM). Boundary condition: $E_z|_{\text{surf}} = 0$.
2. $E_z \equiv 0$: the Transverse Electric Modes (TE). Boundary condition: $\frac{\partial B_z}{\partial n}|_{\text{surf}} = 0$.

For the TE and TM modes this results in an eigenvalue problem for E_z resp. B_z with boundary conditions:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi = -\gamma^2 \psi \text{ with eigenvalues } \gamma^2 := \epsilon\mu\omega^2 - k^2 \quad (9.1.3.14)$$

This has a discrete solution ψ_ℓ with eigenvalue γ_ℓ^2 : $k = \sqrt{\epsilon\mu\omega^2 - \gamma_\ell^2}$. For $\omega < \omega_\ell$, k is imaginary and the wave is damped.

Therefore, ω_ℓ is called the cut-off frequency. In rectangular conductors the following expression can be found for the cut-off frequency for modes $\text{TE}_{m,n}$ or $\text{TM}_{m,n}$:

$$\lambda_\ell = \frac{2}{\sqrt{(m/a)^2 + (n/b)^2}} \quad (9.1.3.15)$$

3. E_z and B_z are zero everywhere for the Transversal Electro-Magnetic Modes (TEM). Then: $k = \pm\omega\sqrt{\epsilon\mu}$ and $v_\ell = v_g$, just as if there were no waveguide. Further $k \in \mathbb{R}$, so no cut-off frequency exists.

In a rectangular, three dimensional resonating cavity with edges a , b and c the possible wave numbers are given by: $k_x = \frac{n_1\pi}{a}$, $k_y = \frac{n_2\pi}{b}$, $k_z = \frac{n_3\pi}{c}$. This results in the possible frequencies $f = vk/2\pi$ in the cavity:

$$f = \frac{v}{2} \sqrt{\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}} \quad (9.1.3.16)$$

For a cubic cavity, with $a = b = c$, the possible number of oscillating modes N_L for longitudinal waves is given by:

$$N_L = \frac{4\pi a^3 f^3}{3v^3} \quad (9.1.3.17)$$

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